

Software and Hardware Tools

1) Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Step 1: Start Wireshark

- Open Wireshark and select your active network interface, such as Wi-Fi or Ethernet.

Step 2: Capture Traffic

- Start the capture and browse Instagram or YouTube for approximately 2 minutes.
- Stop the capture after 2 minutes.

Step 3: Save the Capture

- Use File → Save As and save the capture in .pcap or .pcapng format.

Result

- Top Protocol 1: TCP
- Top Protocol 2: TLS
- Top Protocol 3: DNS

2) Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

Step 1: Test Spotify

- Open Command Prompt on Windows or Terminal on Linux/macOS.
- Run:

```
ping www.spotify.com
```

Step 2: Test Zomato

- Run:

```
ping www.zomato.com
```

Step 3: Record the Results

- Record the response times shown by the commands.
- Use the average response time when it is provided.

Result

Note: Ping times are sample values and can vary by network, location, and time. Use your own command output for final submission.

- Spotify average response time: 45 ms (sample result)
- Zomato average response time: 52 ms (sample result)
- Faster responding site: Spotify (sample result)

3) Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Step 1: Open the Process Viewer

- Windows: Open Task Manager using Ctrl + Shift + Esc.
- macOS: Open Activity Monitor from Applications → Utilities.

Step 2: Find Relevant Processes

- Search for Chrome, WhatsApp Desktop, or another browser-based application that you use.
- Record the process name and memory usage shown.

Result Table

- Process 1 — Name: chrome.exe — Memory: 250 MB (sample result)
- Process 2 — Name: WhatsApp.exe — Memory: 180 MB (sample result)
- Process 3 — Name: chrome.exe — Memory: 120 MB (sample result)
- Process 4 — Name: chrome.exe — Memory: 95 MB (sample result)

4) Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Step 1: Prepare the Scanner

- Install and open Angry IP Scanner or Advanced IP Scanner.
- Only scan your own home or college network, as instructed.

Step 2: Scan the Local Network

- Identify the local IP range used by your Wi-Fi network.
- Start the scan and wait for the results.

Step 3: Record Two Devices

- Select at least two devices from the scan results.
- Record their IP address and device name.

Result

- Device 1 — IP Address: 192.168.1.2 — Device Name: Android Phone (sample result)
- Device 2 — IP Address: 192.168.1.3 — Device Name: Laptop (sample result)

5) Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Hardware Tools

- 1. Cable Tester — Use it to check an Ethernet cable for continuity, incorrect wiring, or damaged pairs.
- 2. Crimping Tool — Use it to attach RJ45 connectors to Ethernet cable when making or repairing network cables.
- 3. Multimeter — Use it to measure electrical values such as voltage or continuity when diagnosing suitable power or hardware-related issues.

Quick Summary

- Cable Tester → Check network cable condition.
- Crimping Tool → Make or repair Ethernet cables.
- Multimeter → Check electrical measurements and continuity.